SSIS

Module 1: SSIS Basics and Package Creation

1. Introduction to SSIS

What is SSIS

SSIS (SQL Server Integration Services) is an ETL tool used to:

- Extract data from multiple sources
- Transform the data
- Load it into destination systems

Key Components

Component	Description
Control Flow	Workflow of tasks
Data Flow	Movement and transformation of data
Connection Manager	Connects to data sources
Variables	Store runtime values
Tasks	Execute operations

SSIS Architecture

1. Source Systems
2. SSIS Package
3. Transformations
4. Destination System

Example Sources

- SQL Server
- Excel
- Flat Files
- APIs

Hands-On Lab 1: Create First SSIS Package

Objective

Create a package that loads data from SQL table to another table.

Steps

1. Open **SQL Server Data Tools**
2. Create **Integration Services Project**
3. Add **Control Flow Task**
4. Add **Data Flow Task**
5. Configure Source and Destination

Example

Source Table

Employees_Source

Destination Table

Employees_Destination

Execute package to copy records.

Processing Flat Files

Flat files include:

- CSV
- TXT
- Log files

Components Used

- Flat File Connection Manager
- Flat File Source
- OLE DB Destination

Hands-On Lab 2: Load CSV File into SQL Server

Example CSV

EmpID,Name,Salary
1,Raj,50000
2,Anita,60000
3,Amit,55000

Steps

1. Create Flat File Connection
2. Define delimiter
3. Configure Data Flow
4. Load into table

Destination Table

Employees

Module 2: Data Handling and Transformations :

Handling Data

Common data operations:

- Data cleansing
- Data conversion
- Filtering

- Aggregation

SSIS Transformations

Common transformations:

Transformation	Purpose
Derived Column	Create new column
Data Conversion	Change datatype
Lookup	Join reference data
Aggregate	Summarize data
Conditional Split	Apply conditions

Hands-On Lab 3: Derived Column Transformation

Input Table

EmpID | Name | Salary

Add new column:

AnnualSalary = Salary * 12

Steps:

1 Add Data Flow Task
2 Add Derived Column Transformation
3 Create new column expression

Salary * 12

Incremental Loading

Incremental loading loads **only new or changed records**.

Benefits:

- Faster ETL
- Less system load

Common approaches:

1 Timestamp
2 ID comparison
3 Change Data Capture

Hands-On Lab 4: Incremental Load

Source Table

Orders_Source

Destination Table

Orders_DWH

Logic

Load only records where OrderDate > LastLoadDate

Use:

- Lookup Transformation
- Conditional Split

MERGE Operation

MERGE combines:

- INSERT
- UPDATE
- DELETE

Example SQL

```
MERGE TargetTable AS T
USING SourceTable AS S
ON T.ID = S.ID

WHEN MATCHED THEN
UPDATE SET T.Name = S.Name

WHEN NOT MATCHED THEN
INSERT (ID, Name)
VALUES (S.ID, S.Name);
```

Fuzzy Lookup

Fuzzy Lookup helps match **similar but not identical data**.

Example

Source	Match
Jon	John
Amit Kumar	Amit K

Used for:

- Data cleansing
- Duplicate detection

Hands-On Lab 5: Fuzzy Lookup

Example

Customer Table

Customer_Name

Reference Table

Master_Customer_Name

Find similar records.

Module 3: Error Handling and Control Flow

Error Handling

Errors occur due to:

- Data type mismatch
- Missing values
- Constraint violations

Methods to handle errors:

- Redirect rows
- Logging
- Error tables

Hands-On Lab 6: Redirect Error Rows

Example

If salary column contains text:

ABC

Instead of failing package:

- Send rows to **Error Table**

SSIS Checkpoints

Checkpoints help restart packages **from failure point**.

Configuration:

1 Enable checkpoints
2 Create checkpoint file
3 Restart package automatically

Pivot Transformation

Pivot converts rows to columns.

Example

Input

Month Sales
Jan 100
Feb 200
Mar 150

Output

Jan Feb Mar
100 200 150

Hands-On Lab 7: Pivot Transformation

Create pivot report for sales data.

Events

Events capture runtime activities.

Examples:

Event	Description
OnError	Error occurs
OnPreExecute	Before execution
OnPostExecute	After execution

Used for:

- Logging
- Alerts
- Auditing

Loops in SSIS

Types of loops:

1 For Loop Container
2 Foreach Loop Container

Example:

Process multiple files in folder.

Hands-On Lab 8: Process Multiple CSV Files

Folder contains

sales_jan.csv
sales_feb.csv
sales_mar.csv

Use **Foreach Loop** to load all files automatically.

Expressions

Expressions allow dynamic values.

Example

Dynamic file path

"C:\\Data\\" + @[User::FileName]

Hands-On Lab 9: Build ETL Pipeline

Scenario

Load customer data from:

CSV → Staging Table → DWH Table

Steps

1 Load CSV
2 Clean data
3 Load to warehouse

Hands-On Lab 10: Database Migration

Source

Legacy Customer DB

Destination

New Data Warehouse

Use:

- OLE DB Source
- Data Flow
- OLE DB Destination

Script Task

Script Task allows **custom code using C#**.

Used when SSIS tasks are insufficient.

Examples

- API call
- File processing
- Custom validation

Example Script

```
public void Main()
{
    MessageBox.Show("SSIS Script Task Executed");
    Dts.TaskResult = (int)ScriptResults.Success;
}
```

Hands-On Lab 11: Script Task

Task

Create script to:

1 Read file name
2 Log execution time
3 Send notification

Final Hands-On Project

Build a complete ETL pipeline:

Scenario

Sales company receives daily files.

Source

CSV Files

Destination

Sales Data Warehouse

Pipeline Steps

1 Load CSV files
2 Clean data
3 Remove duplicates
4 Transform columns
5 Load into warehouse
6 Log errors
7 Archive processed files